

# Spectral Collapse Without Collapse: SVD Dynamics of LoRA Adapters for Low-Resource Turkic Languages

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## Abstract

Spectral collapse—the concentration of adapter weight matrices into a single dominant singular direction—has been proposed as a diagnostic for LoRA training failure. We systematically test this hypothesis by fine-tuning Gemma-2-9B with LoRA on three low-resource Turkic languages (Kyrgyz, Kazakh, Uzbek) and an English control, across eight experimental configurations varying rank ( $r \in \{16, 64\}$ ), learning rate ( $2 \times 10^{-4}$ ,  $5 \times 10^{-4}$ ,  $1 \times 10^{-3}$ ), training duration (3–10 epochs), and cross-lingual transfer (Kazakh  $\rightarrow$  Kyrgyz).

We report three main findings. First, we **challenge the spectral energy threshold hypothesis**: no experiment reaches  $SE > 0.59$  (well below the proposed 0.7 threshold), yet configurations with  $r = 64$  suffer complete functional collapse (NER F1 = 0 across all languages, cross-lingual perplexity  $> 600$ ). Spectral energy actually *decreases* during training as effective rank grows. Second, we identify **Frobenius norm growth** as a more reliable indicator of functional degradation in our setting (Gemma-2-9B, 4-bit quantization): pathological configurations exhibit 15–38 $\times$  norm growth versus only +13% for the best-performing transfer setup. Third, we demonstrate that **sequential Kazakh  $\rightarrow$  Kyrgyz transfer** yields +61% NER F1 improvement (0.253 vs. 0.157) while retaining Kazakh knowledge (perplexity 23.49 vs. 47.58 for direct Kyrgyz training), suggesting that related-language transfer is an effective strategy for extremely low-resource Turkic varieties.

We additionally characterize severe subword fragmentation (3.2 $\times$  vs. English) as a compounding factor for all three languages and release our SVD monitoring toolkit for reproducible spectral analysis of LoRA training dynamics.

## 1 Introduction

Low-Rank Adaptation (Hu et al., 2022) has become the dominant approach for adapting large language models (LLMs) to new domains and languages, particularly when computational resources are limited. By decomposing weight updates into low-rank matrices  $\Delta W = BA$ , LoRA achieves parameter efficiency while preserving most of the pretrained model’s capabilities. However, the internal dynamics of LoRA weight matrices during training—specifically their spectral properties—remain poorly understood, especially for morphologically complex, low-resource languages.

Recent work has identified *spectral collapse* as a potential failure mode of LoRA fine-tuning, where adapter weight matrices degenerate into rank-one approximations dominated by a single singular value (Biderman et al., 2024; Jang et al., 2024). Biderman et al. (2024) proposed that when the spectral energy (SE)—the ratio of the largest singular value to the sum of all singular values—exceeds a threshold of approximately 0.7, the adapter has effectively collapsed and further training becomes unproductive. However, these findings were established primarily on English-language tasks with relatively large training corpora.

Turkic languages present a particularly challenging testbed for LoRA adaptation. Kyrgyz, Kazakh, and Uzbek are agglutinative languages with rich morphological systems, employ both Cyrillic and Latin scripts, and suffer from severe underrepresentation in multilingual LLM pretraining data. Critically, they exhibit extreme subword fragmentation under standard tokenizers (3.2 $\times$  more tokens per word than English), which forces LoRA adapters to make larger weight modifications

to accommodate poorly-covered morphology. These properties make Turkic language adaptation a stress test for parameter-efficient fine-tuning methods.

In this work, we systematically investigate the spectral dynamics of LoRA adapters trained on three Turkic languages, testing two hypotheses:

- **H1 (Spectral Energy Threshold):** Spectral energy exceeding 0.7 predicts functional degradation of LoRA adapters (Biderman et al., 2024).
- **H2 (Cross-Lingual Transfer):** Initializing LoRA adapters from a related-language checkpoint (Kazakh  $\rightarrow$  Kyrgyz) slows spectral degradation and improves downstream task performance.

Our contributions are as follows:

1. We **challenge the SE threshold hypothesis** for low-resource LoRA fine-tuning: maximum SE never exceeds 0.59, yet functional collapse occurs through alternative mechanisms (Section 4.3).
2. We propose **Frobenius norm growth ratio** as a scale-aware diagnostic that reliably separates healthy from pathological training regimes *within underrepresented language settings*, while showing that the same metric is insufficient across language categories (Section 4.4).
3. We demonstrate the **effectiveness of cross-lingual transfer** between related Turkic languages, achieving +61% NER F1 improvement with retained source-language knowledge (Section 4.5).
4. We provide a quantitative **characterization of subword fragmentation** for Kyrgyz, Kazakh, and Uzbek (Section 3.6).
5. We release an open-source **SVD monitoring toolkit** for reproducible spectral analysis of LoRA training.

## 2 Related Work

### 2.1 LoRA and Parameter-Efficient Fine-Tuning

Low-Rank Adaptation (LoRA) (Hu et al., 2022) parameterizes weight updates as  $\Delta W =$

$BA$  where  $B \in \mathbb{R}^{d \times r}$  and  $A \in \mathbb{R}^{r \times k}$  with rank  $r \ll \min(d, k)$ , reducing trainable parameters by orders of magnitude. QLoRA (Dettmers et al., 2023) extends this approach by quantizing the base model to 4-bit precision, enabling fine-tuning of multi-billion parameter models on consumer hardware. Subsequent work has explored rank allocation strategies (Zhang et al., 2023), training stability improvements (Lialin et al., 2023), and theoretical analyses of low-rank update expressiveness (Malladi et al., 2023).

### 2.2 Spectral Analysis of Neural Network Weights

The spectral properties of neural network weight matrices have been studied as indicators of training dynamics and generalization. Martin and Mahoney (2021) demonstrated that well-trained networks exhibit heavy-tailed singular value distributions, a property they termed “implicit self-regularization.” In the context of LoRA, Biderman et al. (2024) showed that LoRA adapters learn less and forget less than full fine-tuning, and identified spectral collapse as a potential diagnostic for training degradation. Jang et al. (2024) provided a more detailed analysis of spectral collapse dynamics, proposing that monitoring the spectral energy ratio can serve as an early warning system for adapter failure. However, both studies focused primarily on English-language settings with relatively large training corpora, leaving open the question of whether these findings generalize to low-resource, morphologically complex languages.

### 2.3 Low-Resource NLP for Turkic Languages

Turkic languages have received increasing attention in the NLP community, though they remain severely underrepresented relative to their speaker populations. Recent benchmarks such as TUMLU (Isbarov et al., 2024) provide standardized evaluation for Turkic language understanding. Work on Kazakh NLP has leveraged Wikipedia and web-crawled corpora (Yeshpanov et al., 2024), while Kyrgyz and Uzbek remain among the least-resourced languages with significant digital presence. The challenge of tokenizer mismatch for agglutinative languages has been documented (Rust

et al., 2021; Ahia et al., 2023), with morphologically rich languages consistently suffering from higher subword fragmentation ratios that inflate sequence lengths and reduce the effective information density per token.

## 2.4 Catastrophic Forgetting in Fine-Tuning

Catastrophic forgetting (McCloskey and Cohen, 1989; Kirkpatrick et al., 2017)—the loss of previously learned knowledge when adapting to new tasks—is a persistent challenge in continual learning. While LoRA’s low-rank constraint was hypothesized to mitigate forgetting by limiting the magnitude of weight updates (Biderman et al., 2024), recent work has shown that aggressive fine-tuning can still induce significant forgetting, particularly for languages distant from the training data (Csaki et al., 2024). The relationship between spectral properties of adapter weights and the degree of catastrophic forgetting has not been systematically studied for multilingual settings.

## 3 Experimental Setup

### 3.1 Base Model

We use Gemma-2-9B (Gemma Team et al., 2024) as our base model, a 9.24 billion parameter decoder-only transformer with 42 layers. The model is quantized to 4-bit precision using the NF4 data type with double quantization (Dettmers et al., 2023), reducing memory requirements to approximately 5.1 billion effective parameters while preserving model quality. All experiments use bfloat16 compute precision. We chose Gemma-2-9B because it possesses baseline linguistic knowledge of Turkic languages from pretraining, allowing us to isolate the spectral dynamics of LoRA *adaptation* from the process of primary language acquisition. The functional collapse observed at  $r=64$  therefore reflects a pathology of the adaptation process, not an inability of the model to represent Turkic tokens.

### 3.2 Training Data

We compile monolingual corpora of approximately 150 MB for each of the three target languages:

- **Kazakh (KZ):** 61,879 records sourced from Kazakh Wikipedia and the sozkz-corpus, totaling 14.1M tokens (151.5 MB).
- **Kyrgyz (KY):** 17,360 records from a curated local corpus comprising literature, history, and encyclopedic texts, totaling 4.4M tokens (151.4 MB).<sup>1</sup>
- **Uzbek (UZ):** 21,242 records from uz-books, Uzbek Wikipedia, and FineWeb-2 with Cyrillic script filtering, totaling 5.4M tokens (152.6 MB). Note that our Uzbek corpus uses Cyrillic script, which reflects a subset of Uzbek digital content; results may not generalize to Latin-script Uzbek, which dominates contemporary formal writing.

Additionally, we compile an **English (EN)** control corpus of 52,268 records from English Wikipedia (155 MB, 12.7M tokens), matched in size to the Turkic corpora.

All texts are chunked into segments of 200–5,000 characters. A 10% held-out split is reserved for validation. The substantial difference in token counts between Kazakh (14.1M) and the other two languages (4.4–5.4M) reflects the higher information density of Kazakh Wikipedia articles; all corpora are matched by raw byte size.

### 3.3 LoRA Configuration

We apply LoRA adapters to all attention and MLP projection matrices: `q_proj`, `k_proj`, `v_proj`, `o_proj`, `gate_proj`, `up_proj`, and `down_proj` (7 modules per layer, 294 total across 42 layers). We set  $\alpha = 2r$  in all experiments and use the paged AdamW optimizer with a cosine learning rate schedule and 5% warmup. Table 1 summarizes all experimental configurations.

### 3.4 SVD Monitoring Protocol

We perform singular value decomposition (SVD) of each `lora_B` weight matrix every 100 training steps, computing five spectral metrics:

<sup>1</sup>Due to licensing restrictions on the source materials, the Kyrgyz corpus cannot be publicly redistributed. Corpus statistics and preprocessing details are provided in Appendix A.

| ID | Experiment     | Language        | Rank | LR                 | Epochs | Trainable % | Init          |
|----|----------------|-----------------|------|--------------------|--------|-------------|---------------|
| E1 | KZ Baseline    | Kazakh          | 16   | $2 \times 10^{-4}$ | 3      | 1.05%       | Random        |
| E2 | UZ Baseline    | Uzbek           | 16   | $2 \times 10^{-4}$ | 3      | 1.05%       | Random        |
| E3 | KY Baseline    | Kyrgyz          | 16   | $2 \times 10^{-4}$ | 3      | 1.05%       | Random        |
| E4 | KY Overfit     | Kyrgyz          | 16   | $2 \times 10^{-4}$ | 10     | 1.05%       | Random        |
| E5 | KY $r=64$      | Kyrgyz          | 64   | $5 \times 10^{-4}$ | 5      | 4.08%       | Random        |
| E6 | KZ→KY Transfer | Kyrgyz          | 16   | $2 \times 10^{-4}$ | 3      | 1.05%       | KZ checkpoint |
| E7 | Diverged       | KY <sup>‡</sup> | 64   | $1 \times 10^{-3}$ | 3      | 4.08%       | Random        |
| E8 | EN Control     | English         | 16   | $2 \times 10^{-4}$ | 3      | 1.05%       | Random        |

Table 1: Summary of experimental configurations. All experiments use Gemma-2-9B with 4-bit quantization,  $\alpha = 2r$ , cosine LR schedule, and effective batch size 16. <sup>‡</sup>E7 was trained on Kyrgyz data but diverged at step 800 due to the aggressive learning rate; it is excluded from downstream evaluation but analyzed in Appendix B.

**Spectral Energy (SE)** quantifies the dominance of the largest singular value:

$$SE = \frac{\sigma_1}{\sum_{i=1}^r \sigma_i} \quad (1)$$

**Effective Rank** (Roy and Vetterli, 2007) measures the number of significantly contributing singular values:

$$\text{EffRank} = \exp \left( - \sum_{i=1}^r p_i \log p_i \right), \quad p_i = \frac{\sigma_i}{\sum_j \sigma_j} \quad (2)$$

**Stable Rank** provides a continuous relaxation of matrix rank:

$$\text{StableRank} = \frac{\|W\|_F^2}{\sigma_1^2} = \frac{\sum_i \sigma_i^2}{\sigma_1^2} \quad (3)$$

**SVD Entropy** measures the uniformity of the singular value distribution:

$$H_{\text{SVD}} = - \sum_{i=1}^r p_i \log_2 p_i \quad (4)$$

**Frobenius Norm** captures the overall magnitude of the weight matrix:

$$\|W\|_F = \sqrt{\sum_{i=1}^r \sigma_i^2} \quad (5)$$

We monitor both `lora_A` and `lora_B` Frobenius norms separately. The spectral collapse threshold is set at  $SE = 0.7$  following Biderman et al. (2024).

### 3.5 Evaluation Protocol

We evaluate each adapter on three tasks:

**Perplexity (PPL):** Computed on held-out validation splits (500 samples, 113–128K tokens per language). We construct a *cross-lingual perplexity matrix* by evaluating each adapter on all three languages, providing a direct measure of catastrophic forgetting.

**Named Entity Recognition (NER):** Using the WikiANN dataset (Pan et al., 2017) with 3-shot prompting on 100 test samples per language. We report micro-averaged F1 across PER, ORG, and LOC entity types.

**Question Answering (QA):** Using TUMLU (Isbarov et al., 2024), a multiple-choice benchmark for Turkic language understanding, with 5-shot log-likelihood evaluation on 695–794 questions per language.

### 3.6 Subword Tokenization Analysis

We measure subword fragmentation as the average number of tokens per whitespace-delimited word for each language. As shown in Table 2, all three Turkic languages exhibit severe fragmentation at approximately  $3.2\times$  the rate of English, reflecting the fundamental mismatch between the Gemma tokenizer (optimized for English and high-resource languages) and agglutinative Turkic morphology.

## 4 Results

### 4.1 Baseline Performance

Table 3 presents the primary evaluation metrics across all experiments. Several patterns emerge from the baseline comparisons.

The Kazakh baseline (E1) achieves the best target-language perplexity (2.73), likely reflecting both the larger effective token count (14.1M vs. 4.4–5.4M) and better representation in Gemma’s pretraining data. The Uzbek

| Language | Tokens/Word | Ratio to EN |
|----------|-------------|-------------|
| English  | 1.15        | 1.00×       |
| Uzbek    | 3.68        | 3.20×       |
| Kyrgyz   | 3.69        | 3.21×       |
| Kazakh   | 3.74        | 3.25×       |

Table 2: Subword fragmentation ratios for the Gemma-2-9B tokenizer. Turkic languages require approximately 3.2× more tokens per word than English, reflecting severe tokenizer mismatch for agglutinative morphology.

baseline (E2) achieves the highest NER F1 on its own language (0.287), though its perplexity (4.03) is higher than Kazakh. The Kyrgyz baseline (E3) shows the weakest overall performance (PPL 4.78, NER F1 0.157), consistent with Kyrgyz being the least resourced of the three languages.

We note apparent cross-lingual NER anomalies: the KZ baseline (E1) achieves higher NER F1 on Uzbek (0.578) than on Kazakh itself (0.164). Wilson confidence intervals at  $n=100$  are wide (e.g., E6 KY NER:  $0.253 \pm 0.085$ , 95% CI [0.168, 0.339]), and the E3 vs. E6 comparison (0.157 vs. 0.253) shows overlapping intervals. We therefore interpret absolute NER F1 values as noisy point estimates and focus our analysis on: (1) the qualitative distinction between functional models ( $F1 > 0$ ) and collapsed ones ( $F1 = 0$ , which is unambiguous), and (2) consistent directional patterns across multiple metrics (PPL, NER, TUMLU) rather than any single number.

The most striking result is the behavior of the  $r=64$  configuration (E5): despite achieving a target-language perplexity of only 5.90 (a modest 23% increase over the baseline), it suffers *complete* NER failure with  $F1 = 0$  across all three languages, driven by 100% parse failures. This indicates that the model has lost its instruction-following capability entirely, a form of functional collapse that perplexity alone fails to capture. We note that this NER failure may reflect loss of structured output formatting rather than loss of entity knowledge per se; however, the TUMLU evaluation, which uses log-likelihood scoring and does not require generation, corroborates the degradation: accuracy drops to 23.2% (near the 25% random baseline for 4-option multi-

ple choice), confirming that the functional collapse extends beyond instruction-following to language understanding itself.

## 4.2 Cross-Lingual Degradation

The cross-lingual perplexity matrix (Table 4) reveals the extent of catastrophic forgetting induced by monolingual fine-tuning.

Several key observations emerge. First, the  $r=64$  configuration (E5) induces catastrophic forgetting: Kazakh PPL rises to 659.25 and Uzbek to 742.73, representing increases of over 240× and 180× relative to the respective baselines. Second, the KZ→KY transfer (E6) retains substantially more Kazakh knowledge than direct Kyrgyz training: KZ PPL of 23.49 versus 47.58 for the KY baseline (a 2× improvement). Third, extended training (10 epochs, E4) nearly doubles cross-lingual PPL relative to the 3-epoch baseline (87.65 vs. 47.58 for KZ; 95.76 vs. 56.85 for UZ), confirming that additional training steps systematically erode non-target-language knowledge. Fourth, the Uzbek baseline (E2) shows the highest asymmetric forgetting: KY PPL rises to 116.89, substantially worse than the reverse direction (UZ PPL under KY training is 56.85), possibly reflecting the greater script distance between Uzbek (historically Latin) and Kyrgyz (Cyrillic).

Critically, the **EN control experiment** (E8) reveals that Frobenius norm growth alone does not predict forgetting. Despite exhibiting 33× norm growth—comparable to the KZ baseline (38×)—the EN adapter induces minimal Turkic language degradation (KZ PPL 5.54 vs. 47.58 for the KY baseline). This demonstrates that forgetting severity depends not only on the magnitude of weight perturbations but also on their *alignment* with pre-trained representations: English weight updates reinforce existing well-represented features, while Turkic updates must restructure underrepresented regions of the weight space, causing disproportionate interference with other languages.

Figure 1 visualizes these patterns as a heatmap.

| ID | Experiment        | Target Lang | PPL (target)      | NER F1 (KY) | NER F1 (KZ) | NER F1 (UZ) | TUMLU (target)     | TUMLU (KZ) |
|----|-------------------|-------------|-------------------|-------------|-------------|-------------|--------------------|------------|
| E1 | KZ Baseline       | KZ          | 2.73              | 0.219       | 0.164       | 0.578       | 38.4%              | 38.4%      |
| E2 | UZ Baseline       | UZ          | 4.03              | 0.138       | 0.167       | 0.287       | 35.4%              | 33.1%      |
| E3 | KY Baseline       | KY          | 4.78              | 0.157       | 0.190       | 0.326       | 34.7%              | 30.2%      |
| E4 | KY Overfit (10ep) | KY          | 4.18              | 0.173       | 0.238       | 0.239       | 33.9%              | 27.8%      |
| E5 | KY $r=64$         | KY          | 5.90              | 0.000       | 0.000       | 0.000       | 23.2%              | 22.2%      |
| E6 | KZ→KY             | KY          | 4.73              | 0.253       | 0.130       | 0.348       | 33.7%              | 31.7%      |
| E8 | EN Control        | EN          | 5.54 <sup>*</sup> | 0.209       | 0.217       | 0.338       | 37.7% <sup>†</sup> | 36.3%      |

Table 3: Main evaluation results across all experiments. PPL is computed on the target language validation set. NER F1 is micro-averaged across entity types using 3-shot WikiANN evaluation (100 samples). TUMLU accuracy is reported for the target language and Kazakh (as a forgetting diagnostic). The  $r=64$  experiment (E5) shows complete NER failure (100/100 parse failures) despite only moderate PPL degradation. <sup>\*</sup>EN Control PPL is reported on Kazakh (the most-resourced Turkic eval set). <sup>†</sup>EN Control TUMLU is reported on Kyrgyz.

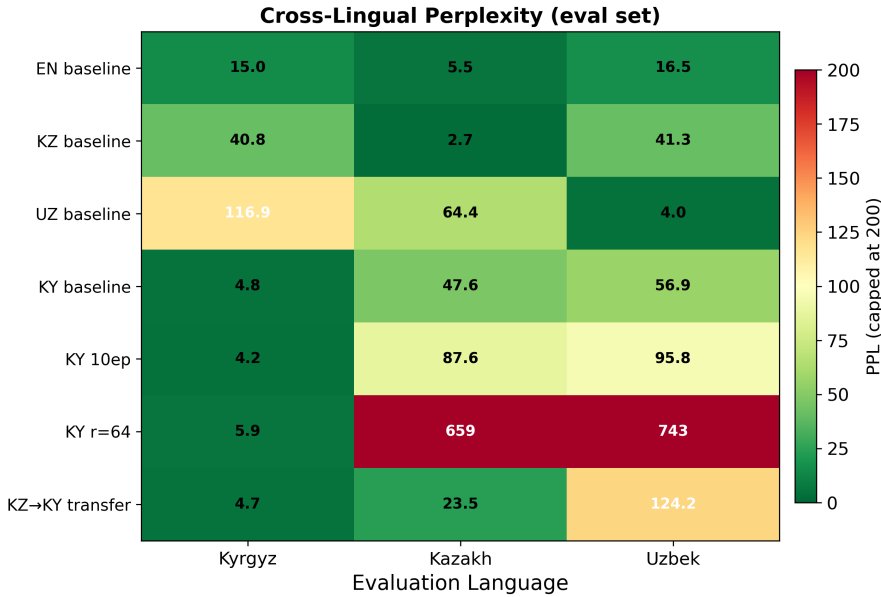


Figure 1: Cross-lingual perplexity heatmap. Darker cells indicate higher perplexity (more forgetting). The  $r=64$  row is clearly pathological, while the KZ→KY transfer row shows the most balanced profile.

### 4.3 Spectral Dynamics: Challenging H1

Our central finding is that the spectral energy threshold of 0.7 proposed by [Biderman et al. \(2024\)](#) is **never reached** in any of our experiments, including those exhibiting severe functional degradation. Figure 2 shows the evolution of mean and maximum SE across training steps for representative experiments.

The KY baseline ( $r=16$ ) shows a mild increase in maximum SE from approximately 0.44 to 0.59 over the course of training, but this remains well below the 0.7 threshold. More strikingly, the pathological  $r=64$  configuration exhibits *decreasing* SE throughout

training, from an initial value of approximately 0.21 to a stable range of 0.10–0.13. This counterintuitive pattern arises because SE is a *shape* metric that measures the relative dominance of the largest singular value: as training progresses, all singular values grow, but the growth is distributed across multiple directions, causing the relative contribution of  $\sigma_1$  to decrease.

The effective rank tells a complementary story: for the  $r=64$  configuration, effective rank *increases* from approximately 6 to 26, indicating that the adapter is utilizing more of its available capacity. Under standard interpretations, increasing effective rank signals healthy training dynamics. Yet this same con-

### SVD Dynamics: Baseline vs Aggressive vs Transfer

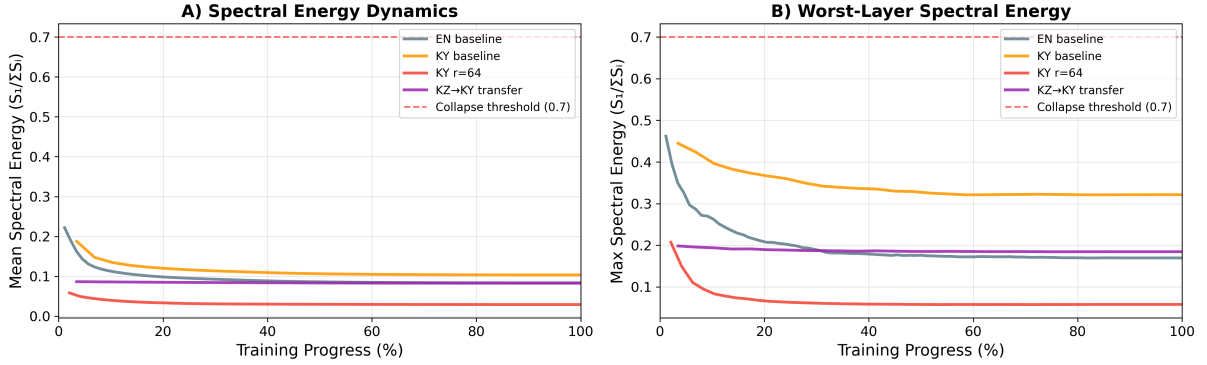


Figure 2: Spectral energy dynamics during training. Top: mean SE across all layers. Bottom: maximum SE across layers. No experiment approaches the proposed 0.7 threshold. The  $r=64$  configuration shows *decreasing* SE despite suffering the worst functional degradation.

| Trained on  | KY PPL      | KZ PPL      | UZ PPL      |
|-------------|-------------|-------------|-------------|
| KZ Baseline | 40.75       | <b>2.73</b> | 41.27       |
| UZ Baseline | 116.89      | 64.44       | <b>4.03</b> |
| KY Baseline | <b>4.78</b> | 47.58       | 56.85       |
| KY Overfit  | <b>4.18</b> | 87.65       | 95.76       |
| KY $r=64$   | 5.90        | 659.25      | 742.73      |
| KZ→KY       | 4.73        | 23.49       | 124.19      |
| EN Control  | 14.96       | 5.54        | 16.55       |

Table 4: Cross-lingual perplexity matrix. Each row is an adapter; each column is the evaluation language. Bold indicates target-language PPL. The  $r=64$  configuration shows catastrophic cross-lingual degradation ( $> 600\times$  PPL increase), while the EN control shows minimal Turkish forgetting despite comparable Frobenius norm growth ( $33\times$ ).

figuration suffers complete NER failure and catastrophic cross-lingual forgetting.

Figure 3 shows the layer-wise SE distribution at the final checkpoint, confirming that no individual layer approaches the collapse threshold.

**H1 is not supported in our setting:** spectral energy below the proposed threshold does *not* guarantee healthy training, and SE can decrease during training that leads to functional collapse. We note that our experiments use 4-bit NF4 quantization, which may itself suppress singular value growth relative to the full-precision setting of Biderman et al. (2024); our findings therefore challenge the generality of the SE threshold rather than refuting it unconditionally. The SE metric fundamentally measures the *shape* of the singular value distribution while ignoring its *scale*—a

| Experiment         | $\ B\ _F^{\text{init}}$ | $\ B\ _F^{\text{final}}$ | Growth       |
|--------------------|-------------------------|--------------------------|--------------|
| KZ→KY Transfer     | 3.043                   | 3.436                    | $1.13\times$ |
| KY Baseline $r=16$ | 0.227                   | 1.899                    | $8.36\times$ |
| EN Control $r=16$  | 0.092                   | 3.076                    | $33.4\times$ |
| KZ Baseline $r=16$ | 0.081                   | 3.107                    | $38.5\times$ |
| KY $r=64$          | 0.566                   | 8.859                    | $15.6\times$ |
| Diverged $r=64$    | —                       | —                        | explodes     |

Table 5: Frobenius norm growth ratios for `lora_B` matrices. EN control and KZ baseline show comparable growth ( $\sim 33$ – $38\times$ ), yet EN produces minimal cross-lingual forgetting (KZ PPL 5.54) while KY baseline with only  $8\times$  growth causes severe forgetting (KZ PPL 47.58). This demonstrates that norm growth alone is insufficient as a universal diagnostic—language representation in the pretrained model modulates the impact of weight perturbations.

critical limitation when diagnosing pathological training regimes in quantized, low-resource settings.

#### 4.4 Frobenius Norm: The Missing Diagnostic

If spectral energy fails as a diagnostic, what metric reliably predicts functional degradation? We find that the **Frobenius norm growth ratio** of the `lora_B` matrices provides a clear separation between healthy and pathological training regimes.

Table 5 and Figure 4 reveal a nuanced picture. The KZ→KY transfer exhibits only  $+13\%$  Frobenius norm growth, while the KY baseline shows  $\sim 8\times$  growth and the pathological  $r=64$  configuration shows  $15$ – $38\times$

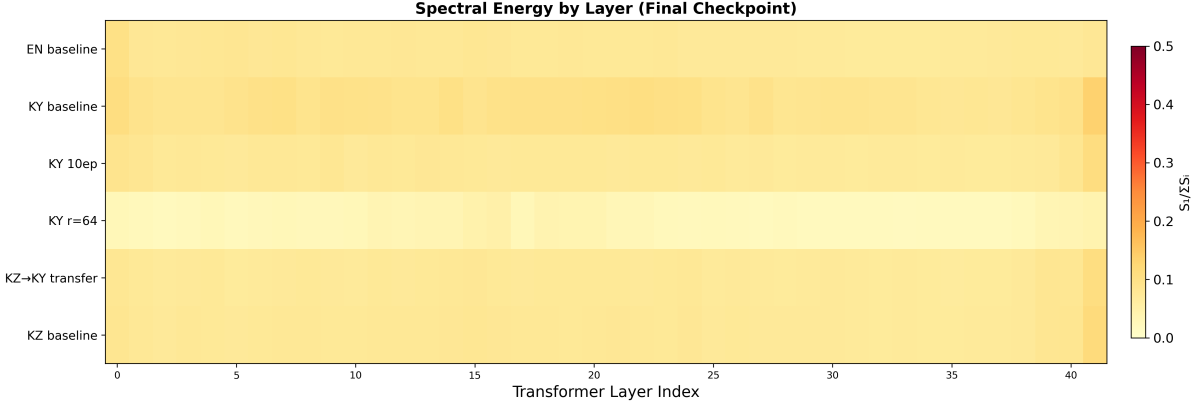


Figure 3: Layer-wise spectral energy at the final training checkpoint. Each row is an experiment; each column is a transformer layer. No layer in any experiment exceeds the 0.7 threshold.

growth. However, the EN control experiment complicates this gradient: despite  $\sim 33\times$  norm growth—comparable to the KZ baseline ( $38\times$ )—EN training causes minimal cross-lingual forgetting (KZ PPL 5.54 vs. 47.58 for KY baseline with only  $8\times$  growth). This reveals that Frobenius norm growth is a reliable diagnostic *within* a language category (underrepresented vs. well-represented) but not *across* categories: the same magnitude of weight perturbation has qualitatively different effects depending on how well the target language is represented in the pretrained model.

Figure 5 visualizes the relationship between Frobenius norm growth and cross-lingual PPL degradation. Within the Turkic experiments, the pattern is monotonically consistent: higher norm growth is associated with worse cross-lingual forgetting ( $\rho_s = 0.26$ ,  $n=6$ , not statistically significant due to small sample size). The EN control, however, is a clear outlier ( $33\times$  growth with minimal forgetting), confirming that norm growth is a necessary but not sufficient condition for catastrophic forgetting. The Frobenius norm is a *scale-aware* metric that captures the total magnitude of weight updates, but the *direction* of those updates—whether they reinforce or disrupt pretrained representations—determines their downstream impact.

#### 4.5 Cross-Lingual Transfer

The KZ→KY transfer experiment (E6) initializes LoRA adapters from the final Kazakh checkpoint (E1) before continuing training on Kyrgyz data. This simple sequential transfer

| Metric           | KY Baseline  | KZ→KY        |
|------------------|--------------|--------------|
| PPL (KY)         | 4.78         | 4.73         |
| PPL (KZ)         | 47.58        | 23.49        |
| PPL (UZ)         | 56.85        | 124.19       |
| NER F1 (KY)      | 0.157        | 0.253        |
| NER F1 (KZ)      | 0.190        | 0.130        |
| NER F1 (UZ)      | 0.326        | 0.348        |
| TUMLU (KY)       | 34.7%        | 33.7%        |
| TUMLU (KZ)       | 30.2%        | 31.7%        |
| $\ B\ _F$ growth | $8.36\times$ | $1.13\times$ |

Table 6: Comparison of direct Kyrgyz training (E3) and Kazakh→Kyrgyz transfer (E6). Transfer improves KY NER F1 by +61% and retains KZ knowledge (PPL 23.49 vs. 47.58). However, UZ PPL increases (124.19 vs. 56.85), suggesting the transfer benefit is primarily between the two closely related Cyrillic-script Kipchak languages.

yields substantial improvements across multiple metrics (Table 6).

The most dramatic improvement is in Kyrgyz NER F1: 0.253 versus 0.157, a relative improvement of +61%. Target-language perplexity is essentially identical (4.73 vs. 4.78), indicating that the transfer does not sacrifice language modeling quality. Kazakh knowledge is substantially better retained: KZ PPL of 23.49 versus 47.58 (a  $2\times$  improvement), and KZ TUMLU accuracy improves from 30.2% to 31.7%.

Interestingly, even Uzbek NER F1 improves slightly under transfer (0.348 vs. 0.326), suggesting some positive transfer to the third Turkic language. However, Uzbek PPL degrades substantially (124.19 vs. 56.85, a  $2.2\times$  increase). This asymmetry has practical implications: sequential transfer between closely re-

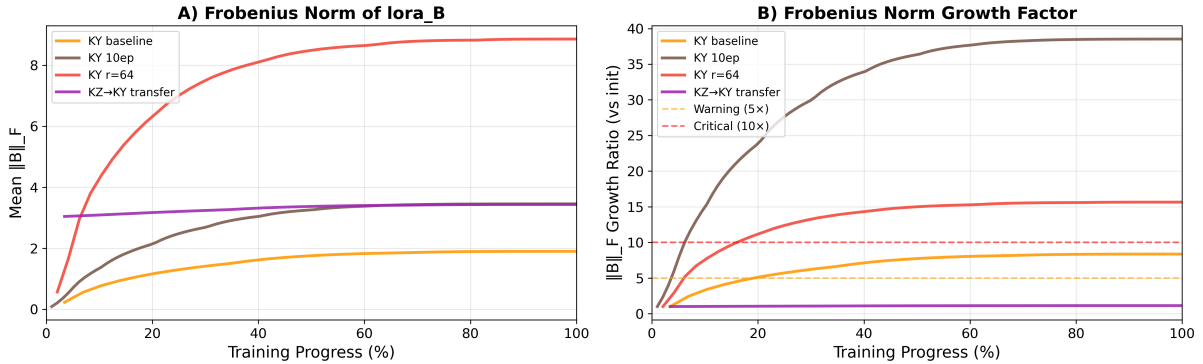


Figure 4: Frobenius norm dynamics of `lora_B` matrices during training. The  $r=64$  configuration shows dramatically larger norm growth compared to  $r=16$  baselines. The transfer experiment begins at the KZ baseline’s final norm and shows minimal additional growth.

lated languages (Kazakh and Kyrgyz are both Kipchak-branch, Cyrillic-script) can *harm* more distant relatives (Uzbek, Karluk-branch, historically Latin-script). The adapter, having been optimized first for Kazakh then for Kyrgyz, specializes its weight updates in the Kipchak direction, displacing representations useful for Uzbek. Practitioners considering chain-of-transfer strategies should therefore monitor not only source-target retention but also collateral damage to typologically distant languages.

**H2 is partially confirmed:** cross-lingual transfer improves downstream task performance and retains source-language knowledge, but the mechanism appears to be *better initialization* rather than *slower degradation*. The transfer adapter begins with an already-trained weight configuration (Frobenius norm growth of only +13% compared to  $8.36\times$  for baseline), requiring minimal additional adaptation to accommodate Kyrgyz data. In effect, related-language pretraining acts as a *spectral regularizer*: by providing a warm-start with already-structured singular value distributions, it constrains subsequent weight updates to a low-perturbation regime that preserves multilingual knowledge.

#### 4.6 The Overfit Paradox

The 10-epoch Kyrgyz experiment (E4) reveals a paradoxical pattern (Figure 6): extended training *improves* target-language perplexity (4.18 vs. 4.78 for the 3-epoch baseline) despite

clear indicators of overfitting (final train loss 0.165 vs. 0.815 for the baseline, indicating severe memorization).

However, this improvement comes at a steep cost in cross-lingual knowledge. Kazakh PPL nearly doubles (87.65 vs. 47.58), Uzbek PPL increases by 68% (95.76 vs. 56.85), and TUMLU accuracy on Kazakh drops from 30.2% to 27.8%. The spectral profile remains healthy throughout all 10 epochs ( $SE < 0.5$ ), further confirming that SE is not a reliable indicator of training pathology.

This pattern suggests a regime of “useful overfitting” for the target language: the model memorizes patterns specific to Kyrgyz text more deeply, yielding better held-out perplexity on in-distribution data, but at the expense of broader multilingual competence. For practitioners focused on a single target language, extended training may be acceptable; for those seeking to maintain multilingual capability, it represents a significant regression.

## 5 Analysis and Discussion

### 5.1 Why Spectral Energy Fails as a Diagnostic

The failure of the SE threshold can be understood through the mathematics of LoRA initialization. Standard LoRA initializes  $A$  with Kaiming uniform and  $B$  with zeros. After the first gradient update,  $B$  acquires small random-scale singular values with roughly uniform magnitudes, yielding  $SE \approx 1/r$ . As training progresses, all singular values grow, but

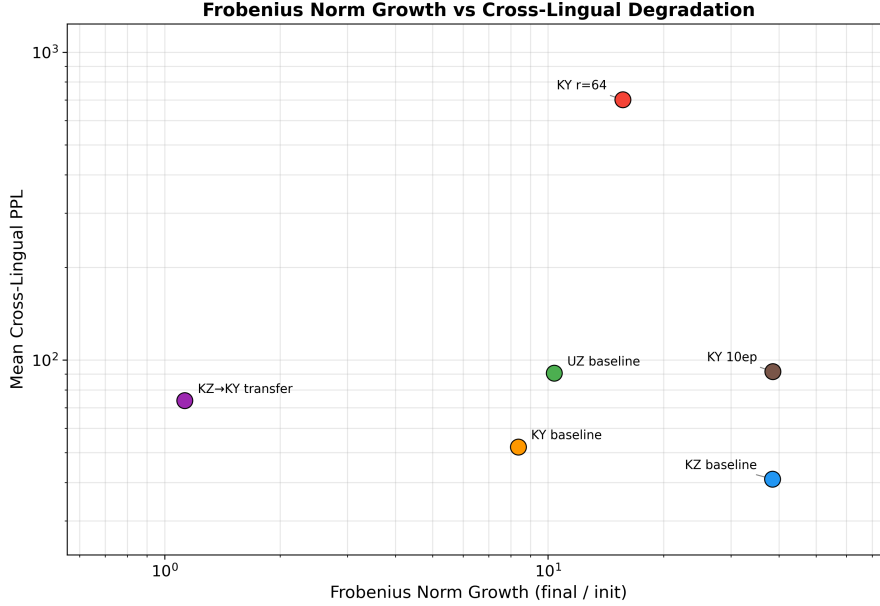


Figure 5: Frobenius norm growth ratio versus cross-lingual perplexity degradation. Larger norm growth is associated with more severe catastrophic forgetting, providing a practical early-stopping criterion.

the growth is typically distributed across multiple directions, causing SE to remain stable or even *decrease*.

For higher-rank configurations ( $r=64$ ), the SE denominator includes more singular values, mechanically driving SE to lower values ( $1/64 \approx 0.016$  at initialization versus  $1/16 = 0.0625$  for  $r=16$ ). This reveals SE as a *rank-biased metric*: since  $SE \approx 1/r$  at initialization, higher-rank adapters *appear* spectrally healthier by definition, creating a dangerous trap for practitioners who monitor SE across configurations with different ranks. A seemingly healthy low-SE value for  $r=64$  may actually correspond to a more pathological state than a higher SE for  $r=16$ .

The fundamental limitation is that SE is a *shape* metric, invariant to rescaling of the singular value spectrum. A weight matrix with singular values  $[100, 50, 25]$  and one with  $[0.001, 0.0005, 0.00025]$  have identical SE, despite representing vastly different magnitudes of weight perturbation. In the low-resource setting, where models must substantially restructure their representations to accommodate poorly-covered languages, the *scale* of these perturbations matters critically.

Figure 7 visualizes the singular value distributions for representative experiments, illustrating how the  $r=64$  configuration devel-

ops a well-distributed but extremely large-magnitude spectrum that SE fails to flag.

## 5.2 Frobenius Norm as a Scale-Aware Alternative

The Frobenius norm,  $\|W\|_F = \sqrt{\sum_i \sigma_i^2}$ , naturally captures the scale of weight perturbations that SE discards. Its squared value equals the sum of squared singular values, providing a direct measure of the total energy in the weight update. The Frobenius norm growth ratio (final/initial) thus quantifies how much the adapter has perturbed the pretrained weights.

Our experiments suggest the following practical thresholds for  $\|B\|_F$  growth monitoring:

- $< 2\times$ : **Healthy** training; minimal forgetting (e.g., transfer setup:  $1.13\times$ ).
- $2\text{--}10\times$ : **Normal** training; moderate forgetting expected (e.g., KY baseline:  $8.36\times$ ).
- $> 10\times$ : **Warning**; significant forgetting likely (e.g.,  $r=64$ :  $15\text{--}38\times$ ).
- Unbounded growth: **Training failure**; divergence imminent (e.g., E7).

We emphasize that these thresholds refer to *relative growth* (ratio of final to initial Frobenius norm) rather than absolute values.



Figure 6: The overfit paradox: 10 epochs of Kyrgyz training (E4) yields better target-language PPL (4.18 vs. 4.78) despite increasing cross-lingual degradation. Train loss continues to decrease while cross-lingual PPL rises monotonically.

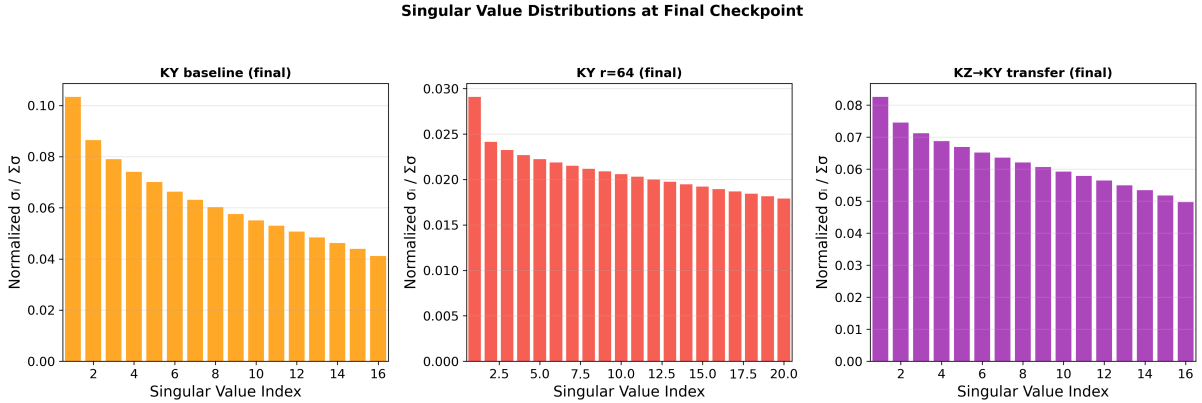


Figure 7: Singular value distributions at the final checkpoint. The  $r=64$  configuration (right) shows a well-distributed spectrum (low SE) with dramatically larger absolute magnitudes compared to the  $r=16$  baseline (left), explaining why SE fails to detect pathological training.

Absolute norms depend on the initialization scheme: standard LoRA (zeros for  $B$ , Kaiming for  $A$ ), PiSSA (Meng et al., 2024), or transfer initialization all produce different baseline magnitudes. The relative growth ratio normalizes this dependency, making it a more portable diagnostic.

**Important caveats:** These thresholds have limited discriminative power in practice. The KZ baseline (E1,  $38.5\times$ ) and EN control (E8,  $33.4\times$ ) both exceed the  $> 10\times$  warning threshold without functional collapse, while the KY baseline (E3,  $8.36\times$ ) falls within the “normal” range yet shows significant cross-lingual forgetting. The  $> 10\times$  threshold is therefore most informative for detecting the extreme  $r=64$  regime; moderate growth (8–

$40\times$ ) in  $r=16$  configurations appears to reflect normal LoRA adaptation for low-resource languages. Additionally, the EN control demonstrates that for well-represented languages, the pretrained model absorbs large weight perturbations without catastrophic interference. We recommend these thresholds as coarse guidelines rather than precise decision boundaries.

### 5.3 Role of Subword Fragmentation

The  $3.2\times$  subword fragmentation ratio observed for all three Turkic languages has several compounding effects on LoRA adaptation:

**Sequence length inflation:** A Kyrgyz sentence of 50 words requires approximately  $50 \times 3.69 = 185$  tokens, versus  $50 \times 1.15 = 58$  tokens for the English equivalent. With a

maximum sequence length of 256 tokens, each training example contains substantially less textual content, reducing the information density of each gradient update.

**Morphological fragmentation:** Agglutinative suffixes that carry grammatical meaning are split across multiple subwords, forcing the model to reconstruct morphological relationships across token boundaries. For instance, a single Kazakh word form may be split into 4-5 subwords, with the root, derivational suffixes, and inflectional endings each occupying separate tokens.

**Link to forgetting severity:** Our EN control experiment (E8) provides direct evidence that subword fragmentation interacts with forgetting dynamics. Despite comparable Frobenius norm growth (EN:  $33\times$  vs. KZ:  $38\times$ ), the EN adapter causes minimal cross-lingual degradation (KZ PPL 5.54) while Turkic adapters cause severe forgetting (KZ PPL 47.58 for KY baseline with only  $8\times$  growth). This suggests that the critical factor is not the *magnitude* of weight updates but their *directionality*: Turkic fine-tuning forces the model to restructure representations in regions poorly covered by pretraining, disrupting shared multilingual features, while EN fine-tuning reinforces already well-represented features with minimal cross-lingual interference.

## 5.4 Practical Recommendations

Based on our findings, we offer the following recommendations for practitioners fine-tuning LLMs on low-resource, morphologically complex languages:

1. **Monitor Frobenius norm growth**, not spectral energy. Set warning thresholds at  $5\times$  growth and stopping thresholds at  $10\times$  growth of  $\|B\|_F$ .
2. **Avoid aggressive rank-LR combinations** for small corpora ( $\sim 150$  MB). The combination of  $r=64$  with  $\text{lr} \geq 5 \times 10^{-4}$  leads to functional collapse in our setting, even when standard spectral metrics appear healthy. Note that E5 confounds rank and learning rate; a controlled  $r=64$ ,  $\text{lr}=2 \times 10^{-4}$  experiment would be needed to isolate the rank effect.

3. **Use related-language transfer** when available. Initializing from a related-language checkpoint provides better weight initialization and substantially improves downstream task performance with minimal additional Frobenius norm growth.
4. **Apply cross-lingual perplexity monitoring** as a forgetting diagnostic. Target-language PPL alone is insufficient; catastrophic forgetting manifests primarily in non-target languages.
5. **Consider early stopping by cross-lingual PPL**, not training loss. Extended training may improve target-language metrics while severely degrading multilingual capability.

## 6 Limitations

Our study has several important limitations. First, all experiments use a single base model (Gemma-2-9B); results may differ for other architectures such as Llama 3 or Mistral, particularly regarding the relationship between rank, norm growth, and functional collapse. Second, while our English control experiment (E8) uses a matched corpus size ( $\sim 150$  MB), differences in data distribution (Wikipedia vs. mixed sources) and tokenizer coverage may partially confound the comparison. Third, NER evaluation relies on few-shot prompting rather than a fine-tuned NER model, introducing noise from instruction-following variation; the complete NER failure for  $r=64$  (100/100 parse failures) may partially reflect loss of instruction-following capability rather than loss of named entity knowledge per se. Fourth, we examine only three of the approximately 35 Turkic languages, excluding Turkish (the highest-resource Turkic language), limiting the generalizability of our cross-lingual transfer findings. Fifth, the fixed corpus size of approximately 150 MB per language precludes scaling analyses that might reveal different dynamics at larger data volumes. Sixth, our SVD monitoring focuses on `lora_B` matrices; `lora_A` dynamics may reveal complementary patterns. Seventh, 4-bit quantization introduces approximation errors that may influence spectral properties in ways not captured by our analysis. Eighth, experiment

E5 ( $r=64$ ) simultaneously increases both rank and learning rate relative to the  $r=16$  baselines ( $5 \times 10^{-4}$  vs.  $2 \times 10^{-4}$ ), confounding the two factors; a controlled  $r=64$ ,  $\text{lr}=2 \times 10^{-4}$  experiment would be needed to isolate the contribution of rank alone. Ninth, our Frobenius norm thresholds are derived from a single experimental setting and should be treated as preliminary guidelines pending validation across diverse models and languages. Finally, our Uzbek corpus is restricted to Cyrillic-script sources; since Uzbekistan officially uses Latin script and most contemporary formal text is Latin-based, our Uzbek results may not generalize to Latin-script data.

## 7 Conclusion

We have presented a systematic study of the spectral dynamics of LoRA adapters during fine-tuning on three low-resource Turkic languages. Our experiments yield three principal findings.

First, the spectral energy threshold hypothesis is **not supported** in our setting: no experiment exceeds SE of 0.59 (well below the proposed 0.7 threshold), yet configurations with  $r=64$  suffer complete functional collapse (NER F1 = 0, cross-lingual PPL > 600). Spectral energy decreases during training as effective rank grows, rendering it an unreliable diagnostic for training pathology.

Second, **Frobenius norm growth ratio** is a useful but language-dependent diagnostic. Within underrepresented languages, it reliably separates healthy from pathological regimes ( $1.13\times$  for transfer vs.  $15\text{--}38\times$  for  $r=64$ ). However, our EN control reveals that well-represented languages can absorb comparable norm growth ( $33\times$ ) with minimal forgetting, demonstrating that the *directionality* of weight perturbations—not just their magnitude—determines forgetting severity.

Third, **sequential cross-lingual transfer** between related Turkic languages is highly effective: Kazakh→Kyrgyz transfer improves NER F1 by +61% (0.253 vs. 0.157) while retaining Kazakh knowledge (PPL 23.49 vs. 47.58), with minimal Frobenius norm growth (+13%).

These findings have immediate practical implications for the growing community of re-

searchers and practitioners adapting LLMs to low-resource languages. Future work should validate our findings across diverse model architectures (particularly in full-precision BF16 to control for quantization effects on spectral dynamics), extend the analysis to additional Turkic languages including Turkish, investigate directional norm regularization as a training objective, and explore adaptive rank selection strategies informed by real-time spectral monitoring. An open question is whether `lora_A` matrices exhibit complementary spectral signals when `lora_B` norms explode—joint analysis of both matrices may yield a more complete diagnostic. Additionally, log-likelihood-based NER evaluation could clarify whether the  $r=64$  configuration loses named entity knowledge entirely or merely loses the instruction-following capability needed for structured output.

## Ethics Statement

This work involves fine-tuning language models on text corpora from three Turkic languages and English. While we aim to improve NLP capabilities for underserved language communities, we acknowledge that language models can amplify biases present in training data. Our training corpora, drawn from Wikipedia, literary texts, and web sources, may contain cultural biases or factual inaccuracies.

**Data availability:** The Kazakh and Uzbek corpora are compiled from publicly available sources (Wikipedia, open book collections). The Kyrgyz corpus is curated from licensed literary and encyclopedic sources and cannot be redistributed due to copyright restrictions. In the spirit of open science, we release all experimental code, SVD monitoring logs, and evaluation results via an anonymous repository.<sup>2</sup> Trained LoRA adapters will be publicly released upon acceptance to preserve author anonymity during review.

## Acknowledgments

We thank the creators of the Kazakh, Kyrgyz, and Uzbek corpora used in this study, as well as the developers of the TUMLU bench-

<sup>2</sup><https://anonymous.4open.science/r/Spectral-Collapse-Turkic-Conference-14D8>

mark for providing standardized Turkic language evaluation resources. All experiments were conducted on personal hardware without external funding.

## Reproducibility Statement

All experimental code (training with SVD monitoring, evaluation, and plotting scripts), SVD monitoring logs, and evaluation reports are publicly available at <https://anonymous.4open.science/r/Spectral-Collapse-Turkic-Conference-14D8>. Fine-tuned LoRA adapters for all eight experiments will be released on a public model hub upon acceptance to preserve author anonymity during review. The Kazakh, Uzbek, and English training corpora are derived from publicly available sources and can be reconstructed using the provided data preparation scripts. The Kyrgyz corpus cannot be redistributed due to licensing restrictions on its source materials; consequently, the KY training stage is not fully reproducible. However, the released KY adapters and SVD logs enable full reproduction of all evaluation and analysis stages.

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## A Detailed Training Configurations

All experiments were conducted on a single NVIDIA GeForce RTX 5080 (16 GB) using PyTorch 2.10, Transformers 5.2, and PEFT 0.18. Table 7 provides the full training statistics.

## B Effective Rank Dynamics

Figure 8 shows the effective rank dynamics across experiments. The  $r=64$  configuration shows increasing effective rank (from  $\sim 6$  to  $\sim 26$ ), indicating growing parameter utilization. The diverged experiment (E7) is the only configuration where effective rank *decreases* (from 43.3 to 37.1), consistent with true spectral collapse.

**E7 divergence analysis.** The diverged experiment ( $r=64$ ,  $\text{lr}=10^{-3}$ ) provides the most extreme datapoint. Training terminated at step 800 due to loss divergence. Even at this point, maximum SE was only 0.18—still far below the 0.7 threshold—while mean Frobenius norm had already reached 6.75 ( $19\times$  growth in just 800 steps, compared to  $15.6\times$  over 4,885 steps for E5). The effective rank decrease ( $43.3 \rightarrow 37.1$ ) in the final 200 steps is the only instance of true rank contraction in our experiments, suggesting that SE-based collapse detection may only trigger *after* the model has already diverged—too late for practical intervention. The Frobenius norm, by contrast, shows clear exponential growth from step 100 onward, providing a much earlier warning signal.

## C Per-Type NER Results

Table 8 breaks down NER performance by entity type for the Kyrgyz evaluation.

## D TUMLU Accuracy by Language

Table 9 presents TUMLU accuracy scores across all languages and experiments.

## E Functional Collapse: Raw Output Examples

To illustrate the nature of functional collapse at  $r=64$ , we present raw model outputs for two tasks. The model completely ignores instructions and exhibits degenerate repetition.

**NER prompt:** Extract named entities from the text. Format each as: ENTITY ||| TYPE

**Input:** Бишкекте Кыргыз Республикасынын Президентин Садыр Жапаров менен Казакстандын Президентин Касым-Жомарт Токаев жолукту.

$r=64$  **output:** Кыргыз Республикасынын Президентинин кызматына кандидат болгон. Кыргыз Республикасынын Президентинин кызматына кандидат болгон. Кыргыз Республикасынын Президентинин кызматына кандидат болгон... (repeats indefinitely)

**Text continuation prompt:** Кыргызстан --- это

$r=64$  **output:** не только, не только, не только, не только... (single bigram repeated 25+ times until max tokens)

Both examples demonstrate that the model has lost instruction-following capability entirely. Rather than extracting entities or generating coherent text, it produces degenerate repetitions—a hallmark of functional collapse distinct from spectral collapse.

| ID | Experiment  | Train Loss | Eval Loss | Eval PPL | Tokens | Time (min) |
|----|-------------|------------|-----------|----------|--------|------------|
| E1 | KZ Baseline | 0.275      | 1.259     | 3.52     | 14.1M  | 477        |
| E2 | UZ Baseline | 1.193      | 1.822     | 6.18     | 5.4M   | 341        |
| E3 | KY Baseline | 0.815      | 1.920     | 6.82     | 4.4M   | 183        |
| E4 | KY Overfit  | 1.050      | 1.963     | 7.12     | 4.4M   | 1193       |
| E5 | KY $r=64$   | 0.274      | 2.164     | 8.71     | 4.4M   | 157        |
| E6 | KZ→KY       | 1.809      | 1.932     | 6.90     | 4.4M   | 355        |
| E8 | EN Control  | 1.821      | 2.017     | 7.51     | 12.7M  | 1351       |

Table 7: Full training statistics for all experiments. Eval loss and PPL are computed on the validation split during training (distinct from the post-hoc evaluation in Table 3). The higher train loss for E6 (KZ→KY) reflects the adapter starting from KZ-optimized weights that initially perform poorly on KY data.

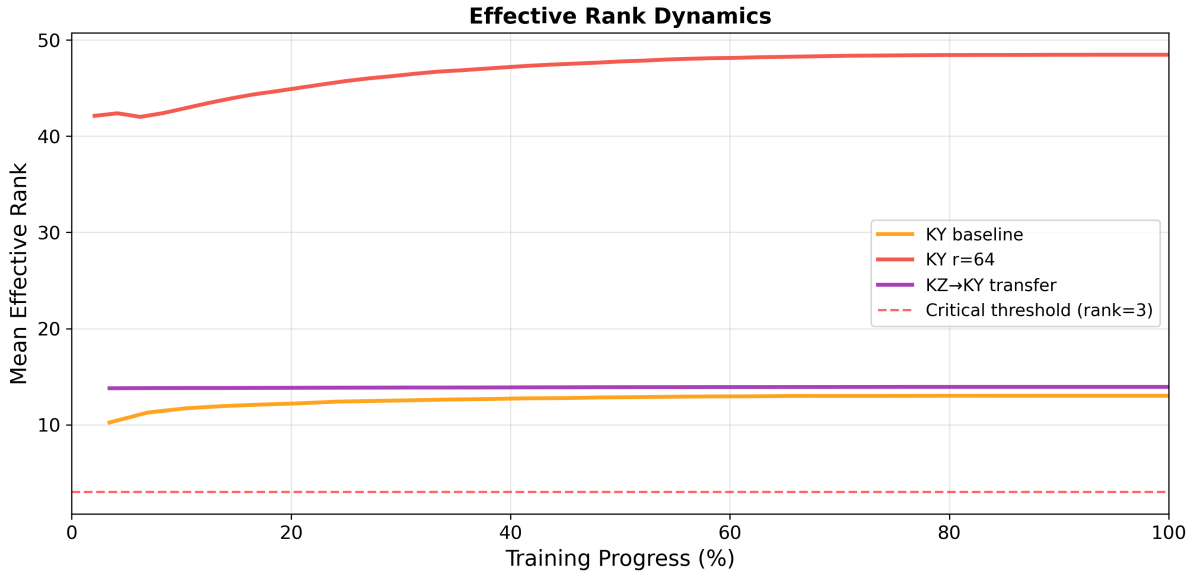


Figure 8: Effective rank dynamics during training. The  $r=64$  configuration shows increasing effective rank despite functional degradation, further evidence that standard spectral metrics can be misleading.

| Experiment          | PER F1 | ORG F1 | LOC F1 |
|---------------------|--------|--------|--------|
| KZ Baseline (on KY) | 0.236  | 0.162  | 0.229  |
| UZ Baseline (on KY) | 0.095  | 0.150  | 0.209  |
| KY Baseline         | 0.101  | 0.160  | 0.266  |
| KY Overfit          | 0.224  | 0.255  | 0.119  |
| KY $r=64$           | 0.000  | 0.000  | 0.000  |
| KZ→KY               | 0.344  | 0.202  | 0.214  |

Table 8: Per-type NER F1 scores on Kyrgyz WikiANN. The KZ→KY transfer achieves the highest PER F1 (0.344), substantially outperforming the KY baseline (0.101). The  $r=64$  configuration produces zero scores across all types due to 100% parse failures.

| Experiment  | KY    | KZ    | UZ    |
|-------------|-------|-------|-------|
| KZ Baseline | 35.7% | 38.4% | 32.8% |
| UZ Baseline | 31.9% | 33.1% | 35.4% |
| KY Baseline | 34.7% | 30.2% | 29.6% |
| KY Overfit  | 33.9% | 27.8% | 31.5% |
| KY $r=64$   | 23.2% | 22.2% | 22.9% |
| KZ→KY       | 33.7% | 31.7% | 30.8% |

Table 9: TUMLU accuracy across all experiments and languages. The KZ baseline achieves the highest accuracy on its target language (38.4%). The  $r=64$  configuration drops to near-random levels ( $\sim 23\%$  on 4-option multiple choice) across all languages.